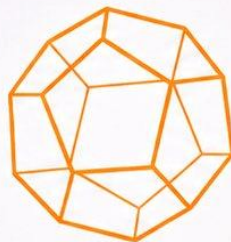




x



Uni AI
Student Initiative

MAKEATHON

— CHALLENGE OWNERS —

The Challenge

- Build a system that turns natural-language questions into dashboard components

"Show me a pie chart of Greek vs English users."

→ Working pie chart. Instantly.

"How is the bot doing this week?"

→ Multi-metric overview. No SQL required.

Works for any question a real ops or product person might ask:

Distributions

Trends

Rankings

Anomaly hunts

Open-ended checks

In English or Greek

What You're Given

conversations.duckdb

90-day synthetic dataset — ~10,000 banking voicebot conversations. Flat SQL-ready views, pre-built for every team.

conversations.jsonl

The same dataset as raw nested JSON. Great for exploratory processing or custom parsing pipelines.

data/schema.md

The column dictionary. The single most important reference document for understanding the data.

data/metrics_dictionary.md

Canonical definitions for containment rate, CSAT, AHT, and all other KPIs. All teams compute the same metrics.

starter/

A small worked example you can crib from or ignore entirely. Optional but useful for a fast start.



Every team works from the exact same pre-built dataset. Do not regenerate or modify it.

From Query to Dashboard

User Query

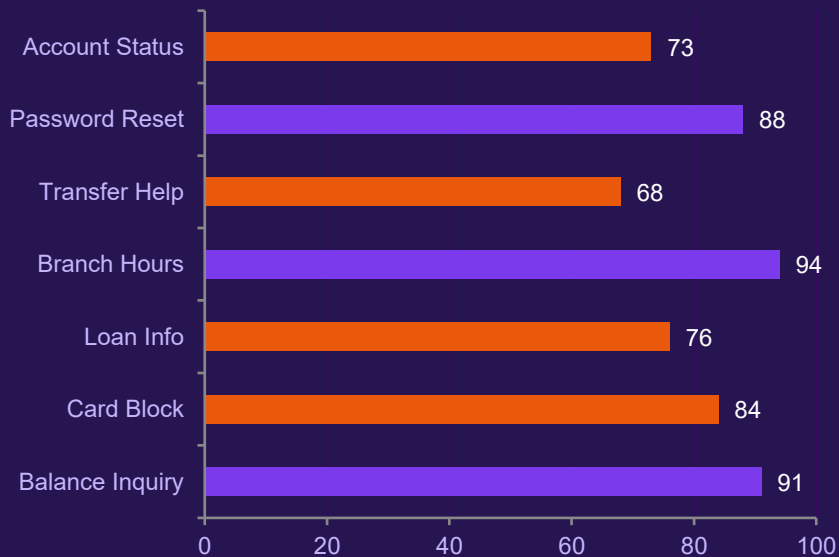
"Show me containment rate by intent type this week"

NR2Dashboard → interprets query → generates SQL → renders chart



Dashboard Output

Containment Rate by Intent — Last 7 Days



What You Build

Anything that takes a natural-language question and produces dashboard components. The form is entirely up to you.

CLI Tool

Jupyter Notebook

Streamlit App

FastAPI + React

Desktop App

Python Script

A live UI is strongly recommended — something that lets a judge type a query and see results in real time wins.

Possible Architectures

Tool-Calling

LLM chooses from `create_pie_chart`, `create_bar_chart`, etc.

✓ *Reliable · Easy to validate · Easy to extend*

Spec Generation

LLM emits Vega-Lite or a similar declarative spec.

✓ *Flexible · Type-safe · Renders cleanly*

SQL + Chart Inference

LLM emits SQL against DuckDB plus a chart type.

✓ *Pairs naturally with the dataset*

Code Generation

LLM emits React/Recharts code, sandbox-rendered.

✓ *Maximum flexibility · Biggest blast radius*

Pick the one your team can ship.

The Dataset

90

days

Synthetic voicebot data

~10K

convos

Banking conversations

2

langs

Greek & English

Key features:

- Nested JSON + flat DuckDB views — pick your preferred access pattern
- Metrics: Containment Rate, CSAT, AHT (Average Handle Time) & more
- Rich enough for distributions, trends, anomaly detection, and rankings
- Pre-built for the event — consistent data for every competing team

Rules & Constraints

ALLOWED

- ✓ Use the dataset as-is — no regeneration, no outside data
- ✓ Respect metric definitions from metrics_dictionary.md
- ✓ All code under a permissive license (MIT / Apache / BSD)
- ✓ Pretrained & API-served LLMs are allowed
- ✓ Pre-trained NL-to-SQL adapters are allowed

NOT ALLOWED

- ✗ Hard-coded lookup tables ("if query contains X → return Y")
- ✗ Modifying or regenerating the dataset
- ✗ Bringing in external data sources

Quick Start

```
git clone https://github.com/SmartRepOrg/makeathon-NR2Dashboard && cd makeathon-NR2Dashboard
```

Good Luck.

Build something you'd actually want to use Monday morning.

Think

Build

Ship